Avionics Start Battery (Mod 6/2315) – General Data

1. General

Mod 6/2315 introduces a 24 volt, 3.3 amp-hour lead-acid avionics start battery. The battery supplies power to the electrical systems required for monitoring engine indications during an engine start. When an engine start is initiated, larger electrical loads are placed on the main battery. This can cause the voltage on the main battery to drop below the minimum voltage required to operate the avionics systems. The avionics start battery is used to maintain the voltage to certain avionics systems so that a start can be accomplished without the loss of engine indications and displays. These systems include MAU1B, MAU2B, DAU1A, DAU1B, DAU2A, DAU2B, AURAL ALERT 1, AURAL ALERT 2, CAUTION LIGHTS, and all four displays. The circuit breakers for these systems are identified with a white circle around the circuit breaker cutouts on the avionics circuit breaker panel.

A schematic of the avionics start battery circuit is shown in Figure 1.

2. Description

A. Avionics Start Battery

The avionics start battery is a 24 volt, 3.3 ampere-hour recombinant gas lead acid type. It is located on the upper of the three avionics racks between the nose baggage compartment and the flight compartment at approximately STA 50.00, beside the MAU. The battery does not require provisions for overboard venting and drains.

B. AVIONICS START BATTERY Switch

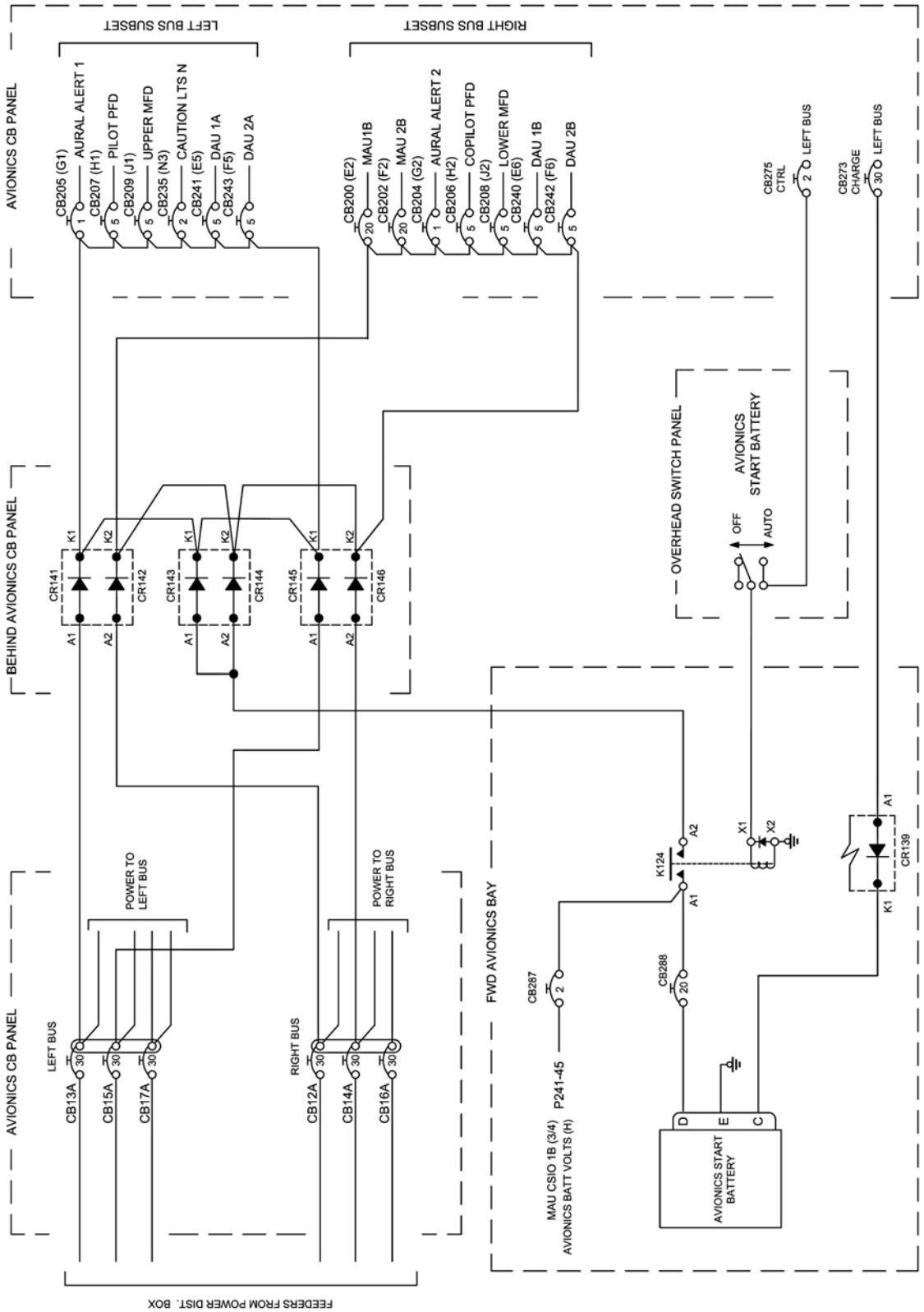
The AVIONICS START BATTERY switch is mounted on the overhead console and is marked, OFF and AUTO. When the switch is set to OFF the avionics start battery output is disconnected. When the switch is set to AUTO, the avionics start battery will maintain voltage to the equipment during the low voltage period, so that engine instruments and warnings continue to function for the engine start.

C. Avionics Start Battery Circuit Breakers

The avionics start battery circuit includes four circuit breakers.

Two of the circuit breakers are located on the top left shelf in the forward avionics bay and are labeled CB287 and CB288. CB287 (2 amps) protects the voltage indication circuit connecting to the MAU. CB288 (20 amps) protects the power output circuit from the battery.

The other two circuit breakers are installed on the avionics circuit breaker panel and are labeled AVIONICS STRT BATT CHARGE (M7) and AVIONICS STRT BATT CTRL (N7). The CHARGE circuit breaker (30 amps) protects the circuit connecting to the battery charge input. The CTRL circuit breaker (2 amps) protects the circuit connecting to the coil of the battery control relay (K124).



Avionics Start Battery Circuit – Electrical Schematic
Figure 1

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D. Avionics Start Battery Circuit Diodes

The avionics start battery circuit includes seven Schottky power diodes. The diodes are packaged in modules containing two diodes per module.

1) Charge Circuit Diode (CR139)

The charge circuit diode is located on the top left shelf in the forward avionics bay and is labeled CR139. The diode is connected to avionics start battery charge input and prevents the avionics start battery from discharging through the charge circuit.

2) Power Isolation Diodes (CR141 thru CR146)

The power isolation diodes are located in the center pedestal forward of the avionics circuit breaker panel and are labeled CR141, CR142, CR143, CR144, CR145, and CR146. The diodes are used to isolate the systems that are not powered by the avionics start battery. The diodes are also used to maintain isolation between the left and right bus.

E. Avionics Start Battery Control Relay

The avionics start battery control relay is located on the top left shelf in the forward avionics bay and is labeled K124. The relay is a SPST type with a continuous duty rating of 50 amps.

3. Operation

The battery is controlled by the AVIONICS START BATTERY switch located on the overhead switch panel. When aircraft power is on and the AVIONICS START BATTERY switch is in the AUTO position, the battery control relay closes and connects the battery in parallel with the main battery. Diodes are used to isolate the battery from the systems that are not required for monitoring engine indications during the start. These diodes are also used to maintain proper isolation between the left and right bus.

The output voltage of avionics start battery is monitored by the AVIONICS BATTERY LOW advisory level CAS message. The CAS message is displayed when the avionics start battery voltage is below 19 volts.

A schematic of the avionics start battery circuit is shown in Figure 1.

4. Removal/Installation

A. Removal of Avionics Start Battery

CAUTION **BEFORE DISCONNECTING THE BATTERY, ENSURE THAT THE FOLLOWING SWITCHES ARE OFF:**

- **AVIONICS START BATTERY SWITCH**
- **MASTER SWITCH (PRE MOD 6/2314)**
- **BATTERY AND EXTERNAL SWITCH (POST MOD 6/2314)**

SUBSTANTIAL LOAD ON THE BATTERY COULD CAUSE DAMAGE TO THE CONNECTOR PINS AND SOCKETS WHEN THE BATTERY IS BEING DISCONNECTED.

Before removing the battery, refer to safety precautions in 24-00-00, Electrical Power – General Data.

1) Set the MASTER switch to OFF (Pre Mod 6/2314); or set BATTERY and EXTERNAL switches to OFF (Post Mod 6/2314).

2) Set the AVIONICS START BATTERY switch to OFF.

3) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Ensure the aircraft power is OFF.

5) Gain access to the forward avionic shelves through the forward baggage compartment.

6) Remove the top cover attached to bulkhead station 37.21.

7) Pull CB273 and CB288 and fit with safety clips.

Note CB273 is located on the avionics circuit breaker panel and is labeled AVIONCIS STRT BATT CHARGE (M7). CB288 is located on the bracket behind the avionics start battery.

8) Disconnect the battery connector (P474).

9) Remove battery mounting screws.

WARNING BATTERIES ARE VERY HEAVY. USE APPROPRIATE LIFTING DEVICES OR EQUIPMENT.

10) Carefully remove battery from aircraft.

B. Installation of Avionics Start Battery

1) Inspect the battery for damage. Cracks in metal or plastic containers are not permitted. Dents in the metal containers that impinge on the interior plastic container are not acceptable.

2) Set the MASTER switch to OFF (Pre Mod 6/2314); or set BATTERY and EXTERNAL switches to OFF (Post Mod 6/2314) and the AVIONICS START BATTERY switch to OFF.

3) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Ensure aircraft power is OFF.

5) Gain access to the forward avionic shelves through the forward baggage compartment.

6) Remove the top cover attached to bulkhead station 37.21.

7) Ensure that avionics shelf is clean and dry.

8) Verify CB273 and CB288 are pulled.

Note CB273 is located on the avionics circuit breaker panel and is labeled AVIONCIS STRT BATT CHARGE (M7). CB288 is located on the bracket behind the avionics start battery.

9) Install battery mounting hardware. Ensure battery is secure on shelf.

10) Connect the battery connector (P474)

11) Reset circuit breakers CB273 and CB288.

12) Re-install the top cover attached to bulkhead station 37.21.

5. Adjustment/Test

A. Capacity Test

Equipment Required:

- Electronic Load, capable of constant current at C1 ADC and 20-28 VDC with EPV termination at 20 VDC
- Mating connector MS27467T15B5P
- Digital Multimeter capable of measuring 18-34 VDC with accuracy of 1% or better

1) Stabilizer the battery at 20°C or higher. The battery must be at this temperature for at least 24 hours.

2) Connect the battery to a digital multimeter using mating connector MS27467T15B5P and measure the open circuit voltage (OCV):

a) If battery is at 20 volts or above, charge the battery following the constant potential charging procedure in Para 6.B.

b) If battery is below 20 volts, remove the battery from the aircraft and do the deep discharge recovery procedure in Para 6.C.

3) Discharge the battery at the C1 rate on the label to the end point voltage (EPV) of 20 volts.

4) Record the time to the end point voltage (EPV).

5) The battery is acceptable for continued use if the ampere hour capacity (actual hours of discharge X ampere rate of discharge) is greater than 85% of the normal capacity (C1) shown on the battery label (i.e. 51 minutes or more).

6) If the battery passes first capacity test, charge the battery as per Para 6.B and return to service.

7) If the battery failed the capacity test for the first time, follow the battery conditioning procedure as per Para 6.D. After battery has been conditioned, repeat the capacity test.

8) If the battery failed the capacity test for a second time, follow the battery conditioning procedure as per Para 6.D. After battery has been conditioned, repeat the capacity test.

9) If the battery failed the capacity test for a third time, or gets very hot during constant charging, replace the battery.

6. Inspection/Check

A. Inspect Avionics Start Battery

A visual inspection in the aircraft should be made of the battery, battery connector and receptacle should be checked for corrosion, contact failure or damage and loose or missing fasteners. Ensure that the identification and information labels are securely attached and legible.

B. Constant Potential Charging Procedure

Equipment Required:

- Constant voltage power supply capable of 28.0 - 28.5 VDC and at least 0.2xC1 ADC output
- Mating connector MS27467T15B5P

Note Constant potential charging is the preferred method of charging the battery. If the battery does not pass the capacity test, the conditioning charge procedure should then be used.

1) Connect the battery to the constant potential charging equipment, using mating connector MS27467T15B5P.

2) Apply a constant potential of 28.25 ± 0.25 volts with a current capacity of at least 0.2xC1 amperes.

Note The output current of the charging equipment should be set as high as possible. RG® series batteries can accept charging rates up to 8xC1.

3) Charge until the charge current falls below 5% of the battery C1 rating (i.e., 0.17 amperes for a 3.3 AH battery) and then continue charging at the same constant potential for a minimum of 4 more hours.

Note Terminate charging if the charge current does not fall below 5% of the battery C1 rating within 8 hours.

C. Deep Discharge Recovery Procedure

WARNING REMOVE BATTERY FROM AIRCRAFT PRIOR TO DOING A DEEP CYCLE CHARGE PROCEDURE. THE PROCEDURE SHOULD BE DONE IN A WELL VENTILATED AREA BECAUSE A SIGNIFICANT AMOUNT OF HYDROGEN GAS MAY BE RELEASED FORM THE BATTERY.

CAUTION IF THE BATTERY BECOMES HOT (ABOVE 55°C/130°F) DURING THE CHARGE, STOP THE CURRENT AND ALLOW THE BATTERY TO COOL TO ROOM TEMPERATURE BEFORE CONTINUING THE CHARGE.

Equipment Required:

- Constant current power supply, capable of C1/10 ADC and at least 34 VDC output
- Mating connector MS27467T15B5P

1) Connect the battery to the constant current charging equipment using mating connector MS27467T15B5P.

2) Charge at a constant current rate of C1/10 (i.e., 0.33 amps for a battery with a C1 rating of 3.3 AH) until the battery voltage reaches 31.0 volts, then continue charging at the same constant current rate for an additional 4 hours. The charge cycle should be continuous when using this method, i.e., no pauses should be included. However, if there is an interruption (e.g., due to a local power outage), continue from where the profile stopped and run to completion.

Note If the battery voltage exceeds 31 volts at the beginning of charge and then drops below 31 volts within 2 hours, continue charging at constant current until the voltage reaches 31 volts a second time. Then continue charging for an additional 4 hours as specified above.

Note If the battery voltage does not reach 31 volts within 24 hours, the charge should be terminated.

3) After the charge is complete, allow the battery to cool down for at least 8 hours or until the battery temperature is within 10°C/18°F of the ambient temperature, then proceed with the capacity test.

D. Battery Conditioning Procedure

WARNING REMOVE BATTERY FROM AIRCRAFT PRIOR TO DOING A BATTERY CONDITIONING PROCEDURE. THE PROCEDURE SHOULD BE DONE IN A WELL VENTILATED AREA BECAUSE A SIGNIFICANT AMOUNT OF HYDROGEN GAS MAY BE RELEASED FORM THE BATTERY.

CAUTION A CONDITIONING CHARGE IS ONLY NECESSARY IF THE BATTERY FAILS THE CAPACITY TEST. THIS PROCEDURE MAY SHORTEN THE BATTERY'S LIFE IS DONE ON A REPETITIVE BASIS.

CAUTION IF THE BATTERY BECOMES HOT (ABOVE 55°C/130°F) DURING THE CONDITIONING CHARGE, STOP THE CURRENT AND ALLOW THE BATTERY TO COOL TO ROOM TEMPERATURE BEFORE CONTINUING THE CHARGE.

Equipment Required:

- Electronic Load, capable of constant current at C1 ADC and 20-28 VDC with EPV termination at 20 VDC
- Constant Current Power Supply, capable of C1/10 ADC and at least 34 VDC output
- Mating connector MS27467T15B5P

1) Remove battery from aircraft as per Para 4.A.

2) Connect battery to electronic load using mating connector MS27467T15B5P.

3) Discharge battery at C1 rate, noted on the battery label, to end point voltage of 18 volts.

4) Charge at a constant current rate of C1/10 for 16 hours and then discontinue charging. The conditioning charge does not need to be continuous, i.e., pauses may be included. For example, an 8 hour charge on the first day followed by an 8 hour charge on the second day is acceptable.

5) Allow the battery to cool down for at least 8 hours or until the battery temperature is within 10°C/18°F of the ambient temperature, then proceed with the capacity test as per Para 5.A.

E. Check Avionics Battery Diodes

Note Refer to the PSM 1-64-2W Wiring Diagrams manual to familiarize with the circuitry.

Equipment Required:

- Digital multimeter
- Torque driver

Note Diodes that do not pass this check must be replaced.

1) Charge Circuit Diode (CR139)

a) Ensure aircraft power is OFF.

b) Gain access to the forward avionic shelves through the forward baggage compartment.

c) Remove the top cover attached to bulkhead station 37.21.

d) Pull CB273 and CB288 and fit with safety clips.

Note CB273 is located on the avionics circuit breaker panel and is labeled AVIONCIS STRT BATT CHARGE (M7). CB288 is located on the bracket behind the avionics start battery.

e) Disconnect battery connector (P474).

f) Using a digital multimeter in diode test mode, measure CR139 and confirm forward-bias reading is approximately 0.25V and reverse-bias reading is open.

g) Reconnect battery connector (P474).

h) Reset CB273 and CB288 circuit breakers.

i) Reinstall the top cover attached to bulkhead station 37.21.

2) Power Isolation Diodes (CR141 thru CR146)

Refer to Figure 2 for terminal arrangement of diodes.

a) Ensure aircraft power is OFF.

b) Remove center pedestal cover to gain access to diodes behind avionics circuit breaker panel.

c) Disconnect wire 1P128A14 from CR141 terminal A1.

d) Using a digital multimeter in diode test mode, measure CR141 and CR145 and confirm forward-bias reading is approximately 0.25V and reverse-bias reading is open.

e) Reconnect wire 1P128A14 to CR141 termianl A1. Torque terminal screw to 11.5 - 12.5 in-lbs.

CAUTION **DO NOT OVER TIGHTEN SCREWS. THIS CAN DO DAMAGE TO THE DIODE MODULE.**

f) Disconnect wire 2P128A14 from CR142 terminal A1.

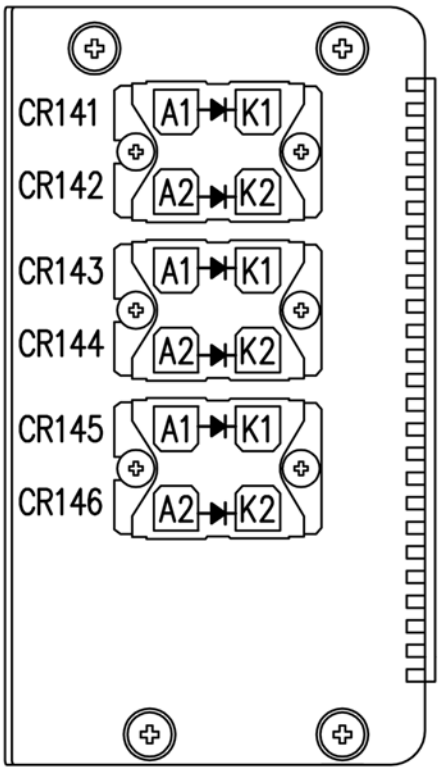
g) Using a digital multimeter in diode test mode, measure CR142 and CR146 and confirm forward-bias reading is approximately 0.25V and reverse-bias reading is open.

h) Reconnect wire 2P128A14 to CR142 terminal A1. Torque terminal screw to 11.5 - 12.5 in-lbs.

CAUTION **DO NOT OVER TIGHTEN SCREWS. THIS CAN DO DAMAGE TO THE DIODE MODULE.**

i) Using a digital multimeter in diode test mode, measure CR143 and CR144 and confirm forward-bias reading is approximately 0.25V and reverse-bias reading is open.

j) Re-install center pedestal cover.



CR141 thru CR146 Diode Terminal Arrangement
Figure 2

7. Troubleshooting

SYMPTOM	POSSIBLE CAUSE	CHECK THAT	REMEDIAL ACTION
1) Battery does not hold charge.	Battery beyond serviceable life.		Replace battery.
2) Battery capacity less than 85% of rating.	Battery beyond serviceable life.		Replace battery.
3) Battery gets hot during constant potential charging.	Battery beyond serviceable life.		Replace battery.
4) Battery not charging in aircraft.	Faulty charge circuit diode.	Check CR139 in accordance with Para 6.E.1).	Replace faulty diode.
	Battery beyond serviceable life.	Do capacity test in accordance with Para 5.A.	Replace battery.
5) Power interruption to displays during start with fully charged avionics start battery.	Battery output circuit breaker (CB288) tripped.		Reset circuit breaker.
	Faulty power isolation diodes.	Check CR141 thru CR146 in accordance with Para 6.E.2).	Replace faulty diodes.
	Faulty battery control relay.	Relay does not close when AVIONICS START BATTERY switch is in AUTO position.	Replace faulty relay.

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